



Software Quality Frameworks & ISO-9126

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What is Software Quality?

Software Quality is the degree to which a software system satisfies its explicit functional requirements and implicit user expectations. Correctness and defects are its primary metrics.

Core Perspectives of Quality

- **User Perspective:** Reliability, usability, and responsiveness during application execution.
- **Developer Perspective:** Maintainability, clear structure, and testability of the source code.
- **Business Perspective:** Cost-effectiveness, time-to-market, and alignment with contract specs.
- **Correctness vs. Defects:** Correctness is adherence to specs; defects are deviations (faults, errors, failures) in execution.

Historical Defect Measurements

- **Defect Density:** Measured as the number of defects per Thousand Lines of Code (KLOC):
$$\text{Density} = \text{Defects} / \text{KLOC}.$$
- **Fault vs. Failure:** A 'fault' is a bug in the code. A 'failure' is the execution of a fault causing incorrect user results.
- **Historical Progression:** Quality has progressed from pure post-development testing to active quality engineering throughout the development lifecycle.

ISO-9126 Quality Model

ISO-9126 is an international standard defining a hierarchical software quality framework:

The Six Core Quality Characteristics

Characteristic	Definition / Focus	Key Metric Focus
Functionality	Existence of functions matching stated and implicit needs.	Accuracy, interoperability, compliance, security.
Reliability	Software's capability to maintain performance levels over time.	Maturity, fault tolerance, recoverability.
Usability	Ease of understanding, learning, and operating the system.	Understandability, learnability, operability.
Efficiency	Performance relative to CPU, memory, and database resources.	Time behavior, resource utilization.
Maintainability	Effort needed to make specified code modifications.	Analyzability, changeability, stability, testability.

Comparison of Quality Models

A structural comparison of ISO-9126 with McCall's and Boehm's quality frameworks:

Evaluation Category	McCall's Quality Model	Boehm's Quality Model	ISO-9126 Standard
Structure	3 product pillars (Operations, Revision, Transition)	Hierarchical tree representing user needs	6 main characteristics, 27 sub-characteristics
Focus Area	Code properties and maintainer ease	Hardware utilization and efficiency	External quality, internal quality, quality in use
Portability	Pillar of Product Transition	Classified under General Utility	Primary top-level characteristic
Target Audience	Software maintainers & developers	System engineers & customers	End-users, developers, and QA auditors

Applications of Quality Frameworks

How software companies implement these quality models in practice:

Contract Requirements Alignment

Organizations use ISO-9126 terms to specify contractual SLA bounds. E.g., 'Reliability must guarantee 99.9% uptime (fault tolerance) and Recoverability must be within 1 hour.'

Automated Code Scanning

Static analysis tools (e.g. SonarQube) map code issues to Maintainability, Security, and Reliability buckets, calculating 'Technical Debt' dynamically.

Vendor Benchmarking

Allows IT enterprises to objectively evaluate third-party software products against standardized quality checklists before procurement.

Safety-Critical Certifications

Aerospace and medical software must demonstrate compliance with strict Reliability and Maintainability standards prior to public deployment.



THANK YOU

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